 Name________________________

The Fundamental Theorem of Algebra

Date________________ Period____

State the number of complex zeros, the possible number of real and imaginary zeros, the possible number of positive and negative zeros, and the possible rational zeros for each function.

1) $f(x) = 5x^4 - 36x^2 - 81$

2) $f(x) = 15x^5 + 3x^4 + 140x^3 + 28x^2 + 45x + 9$

3) $f(x) = 5x^3 - x^2 - 5x + 1$

4) $f(x) = 3x^3 + 11x^2 + 5x - 3$

5) $f(x) = 10x^5 - 15x^4 + 12x^3 - 18x^2 + 2x - 3$

6) $f(x) = 5x^5 - 25x^4 + 46x^3 - 230x^2 + 9x - 45$

State the possible rational zeros and an interval in which all real zeros lie for each function. Then factor each to linear and irreducible quadratic factors.

7) $f(x) = 2x^4 - 11x^2 + 9$

8) $f(x) = 27x^3 + 1$